ML PROJECT 2: APPLYING MLP

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THE PROJECT SCOPE

- Using the **UCI Adult Income** dataset (also known as the Census Income dataset) to predict whether a person earns more than $50K/year.
- **Train an MLP** to classify income as **1** (>50k) and **-1** (anything else)
  - **MLPs (Multi Layer Perceptron)** are a network of neurons (nodes) arranged in layers that can learn patterns from data. It learns by adjusting internal weights over many examples.

THE DATA

Cleaning and Preprocessing

Numeric Columns:

Unnamed: 0, age, fnlwgt, education-num, capital-gain,
Capital-loss, hours-per-week

Cleaning Numeric Columns:

- Filled in any missing numbers with the column's median value.
- Scaled the numbers so everything is on a similar range.

Categorical Columns:

workclass, education, marital-status, occupation, relationship,
race, sex, native-country

Cleaning Categorical Columns:

- Filled in missing categories with the most common value.
- Converted text labels into numeric "one-hot" columns.

BASELINE MODEL

- **Two hidden layers:** 64 and 32 neurons
- **Activation:** ReLU — enables nonlinear learning
- **Optimizer:** Adam — adaptive and efficient convergence
- **Training limit:** up to 500 iterations
- **Early stopping:** stops if no improvement after 50 iterations
- **Validation fraction:** 10% of training data for model tuning
- **Random state fixed** for reproducibility

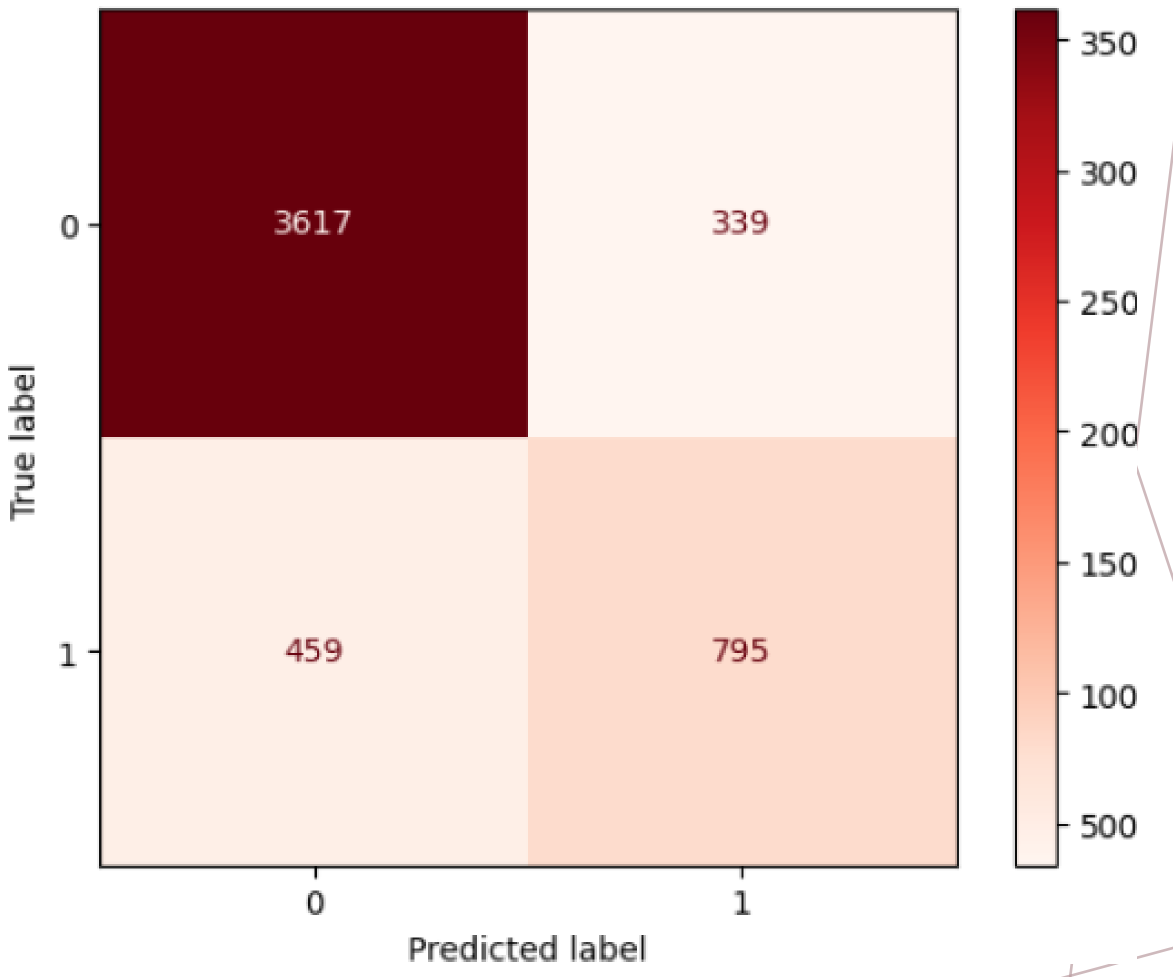
BASELINE MODEL

Classification Report:

	precision	recall	f1-score	support
-1	0.8874	0.9143	0.9006	3956
1	0.7011	0.6340	0.6658	1254
accuracy			0.8468	5210
macro avg	0.7942	0.7741	0.7832	5210
weighted avg	0.8425	0.8468	0.8441	5210

Baseline Accuracy: 0.8468
Baseline Balanced Accuracy: 0.7741
Baseline F1 Score: 0.6658

Train Accuracy: 0.8697
Test Accuracy: 0.8468



MODEL TUNING: Using MLP hyperparameter search (GridSearchCV)

Hidden layer options: (32,), (64,), (64, 32), (128,), (128, 64)

Activations tested: ReLU and Tanh

Regularization (α): 0.0001 to 0.005 (L2 penalty)

Learning rates: 0.001e and 0.0005

Training setup: Early stopping and 3-fold stratified cross-validation

Scoring metric: Balanced accuracy — fairer for imbalanced data

Refits best model on full training data for final evaluation

MODEL TUNING: Using MLP hyperparameter search (GridSearchCV)

Top 5 configurations for balanced accuracy:

	mean_test_score	param_clf__hidden_layer_sizes	param_clf__activation	param_clf__alpha	param_clf__learning_rate_init
48	0.782507	(128, 64)	tanh	0.0010	0.0010
17	0.778646	(128,)	relu	0.0010	0.0005
7	0.778351	(128,)	relu	0.0001	0.0005
54	0.776351	(64, 32)	tanh	0.0050	0.0010
34	0.776093	(64, 32)	tanh	0.0001	0.0010

TUNED MODEL

- **Activation:** tanh
- **Hidden Layers:** (128, 64)
- **Regularization (α):** 0.001
- **Learning Rate:** 0.001
- **Optimizer:** Adam
- **Training Limit:** 500 iterations
- **Early Stopping:** Enabled (stops after 10 no-improvement rounds)
- **Validation Fraction:** 0.10
- **Random State**: Fixed for reproducibility

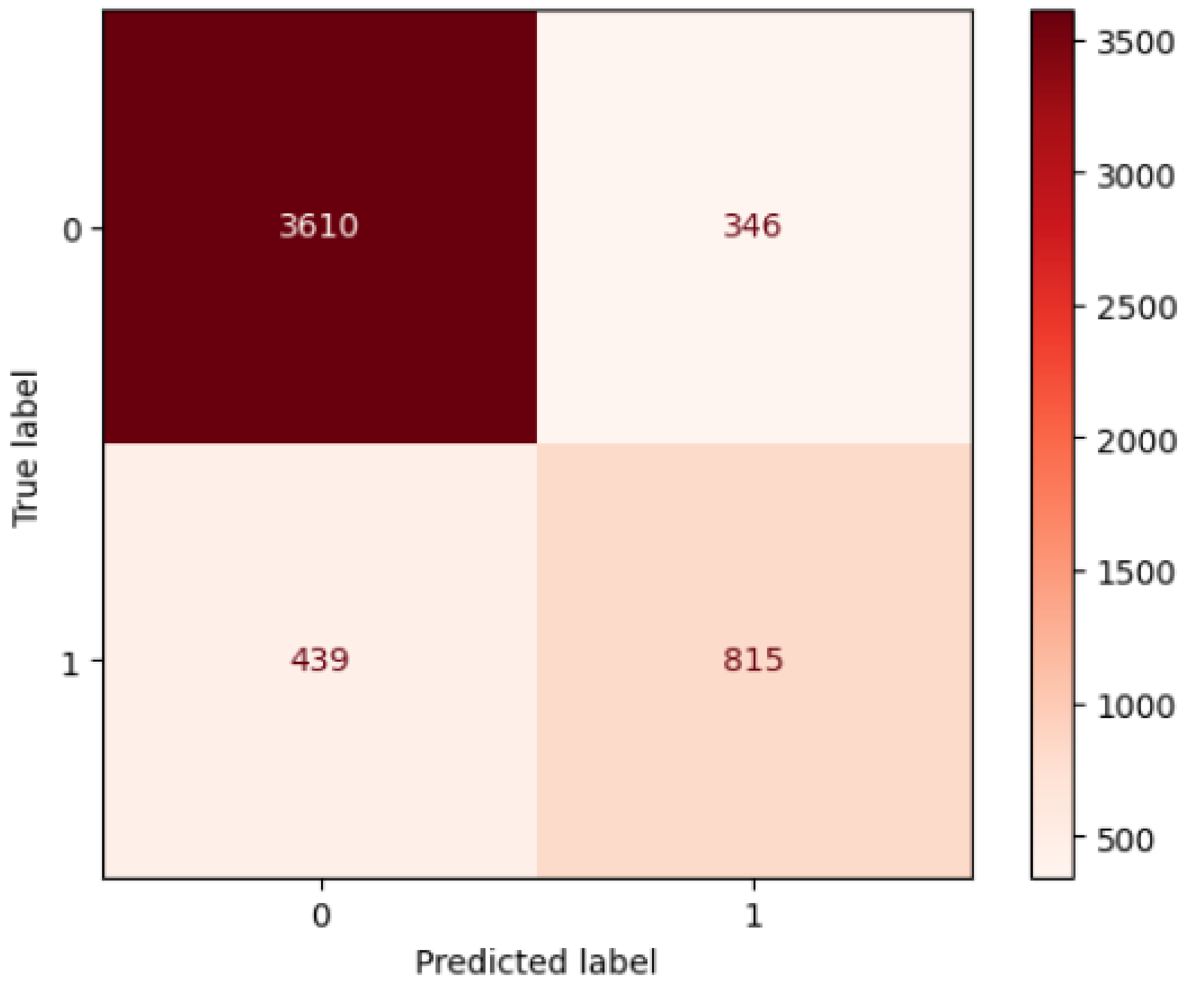
TUNED MODEL

```
Classification Report:
              precision    recall  f1-score   support

          -1     0.8874    0.9143    0.9006      3956
           1     0.7011    0.6340    0.6658      1254

    accuracy                         0.8468      5210
   macro avg     0.7942    0.7741    0.7832      5210
weighted avg     0.8425    0.8468    0.8441      5210
```

```
Baseline Accuracy:          0.8468
Baseline Balanced Accuracy: 0.7741
Baseline F1 Score:          0.6658
```

```
Train Accuracy (best): 0.8549
Test  Accuracy (best): 0.8493
```

```
Best threshold for F1: 0.400  |  F1 at best threshold: 0.6898
```



REFLECTION AND CONCEPTUAL ANSWERS

Why did we choose this MLP architecture?

- **(64, 32) layers:** Simple, fast, and strong for tabular data.
- **ReLU activation**: Trains quickly and works well with sparse/one-hot features.
- **Early stopping:** Halts when validation performance levels off which prevents overfitting and saves time.
- **Outcome:** Stable, efficient training without chasing tiny, costly improvements.

REFLECTION AND CONCEPTUAL ANSWERS

How Did We Monitor and Mitigate Overfitting?

- **Early stopping:** Validation split stops training when progress plateaus.
- **Stratified hold-out:** Train/test split reveals gaps between train and test performance.
- **Imbalance-aware metrics:** Used balanced accuracy and F1 instead of plain accuracy.
- **Light regularization:** Small network + L2 penalty to control variance.

REFLECTION AND CONCEPTUAL ANSWERS

What Ethical Concerns Might Arise with this Dataset/Model?

- **Historical bias:** Features like education or occupation may reflect social inequities that could be amplified by our model.
- **Sensitive inferences:** Income predictions can affect opportunities so misuse of this income information may harm groups.
- **Transparency & governance:** Clearly communicate limits, avoid *proxy discrimination*, audit subgroup performance.
  - A model may unintentionally favor or harm groups by using features that are highly correlated with sensitive attributes.

REFLECTION AND CONCEPTUAL ANSWERS

Why are Activation Functions Necessary in Neural Networks?

- **Add nonlinearity**: Enable learning complex, curved decision boundaries.
- **Make depth useful:** Without them, multiple layers act like a single linear model.
- **Practical choices:**
  - ReLU: Fast, avoids vanishing gradients but has the most hidden layers.
  - Tanh: Zero-centered, useful in smaller or specialized networks.
  - Sigmoid: Probability output for binary tasks.
  - Softmax: Multi-class probability outputs.